

5. PROJECT DEVELOPMENT PHASE

Medical Diagnosis Center by Service Portal (ServiceNow)

The objective of the development phase is to implement the required Medical Diagnosis Center activities in ServiceNow and demonstrate the complete application flow. The development includes application creation, patient registration, diagnostic test management, appointment booking, appointment cancellation, refund calculation, report generation, and Service Portal access.

Development Flow

ServiceNow Application → Patient Registration → Diagnostic Test Management → Test Selection → Appointment Booking → Appointment Management → Appointment Cancellation → Refund Calculation → Appointment Completion → Automatic Report Generation → Report Management → Service Portal Access

5.1 Phase 1 — Application Creation

Create the Medical Diagnosis Center application in ServiceNow.

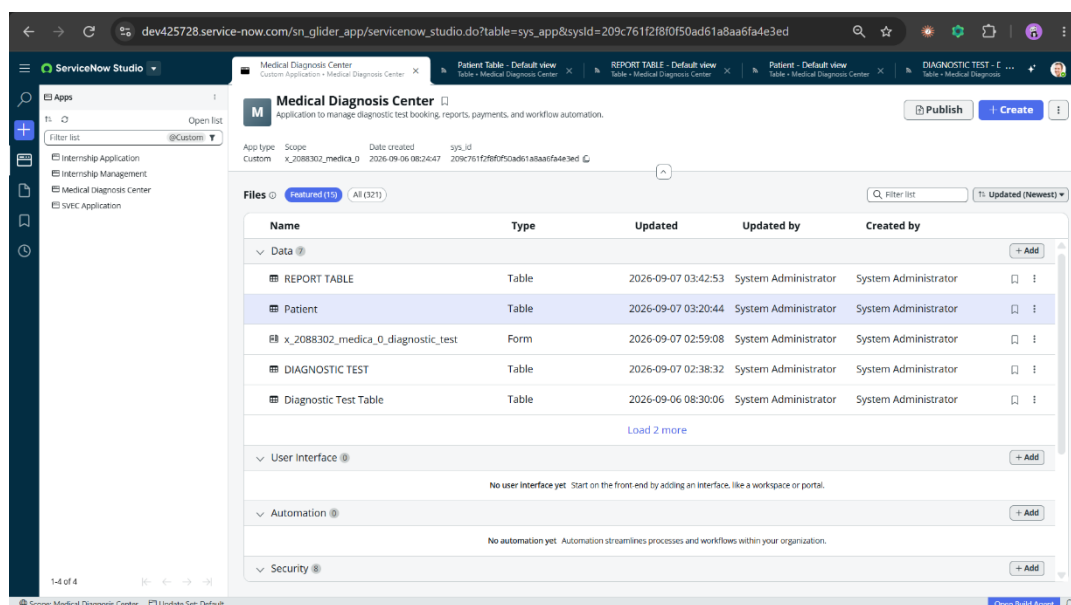
Implementation

Create the application in ServiceNow and configure the required application components for the Medical Diagnosis Center. The application provides the base environment for managing patients, diagnostic tests, appointments, and reports.

Result

The Medical Diagnosis Center application is created and ready for development.

Figure 6: Medical Diagnosis Center ServiceNow Application



5.2 Phase 2 — Patient Table Development

Create the Patient table to store and manage patient information.

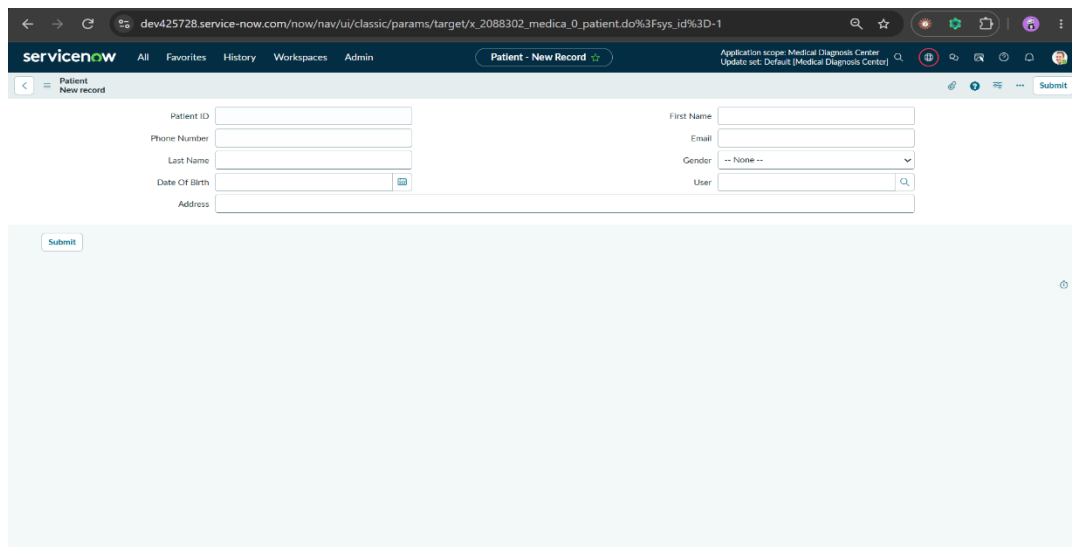
Implementation

Create the Patient table and configure the required fields for patient registration and management. The table is used to maintain patient records that can be associated with appointments.

Result

The Patient table is created and patient information can be maintained in the application.

Figure 7: Patient Table



The screenshot shows a web browser window with the URL `dev425728.service-now.com/now/nav/ui/classic/params/target/x_2088302_medica_0_patient.do%3Fsys_id%3D-1`. The page title is "Patient - New Record". The form contains the following fields:

- Patient ID
- Phone Number
- Last Name
- Date Of Birth
- Address
- First Name
- Email
- Gender (dropdown menu with "-- None --" selected)
- User (lookup field)

A "Submit" button is located at the bottom left of the form.

5.3 Phase 3 — Diagnostic Test Table Development

Create the Diagnostic Test table to maintain the diagnostic tests provided by the center.

Implementation

Create the Diagnostic Test table and configure the required fields for maintaining diagnostic test information. The test records are used when patients select a diagnostic test for an appointment.

Result

The Diagnostic Test table is created and diagnostic test information can be maintained.

Figure 8: Diagnostic Test Table

dev425728.service-now.com/now/nav/ui/classic/params/target/x_2088302_medica_0_diagnostic_test.do%3Fsys_id%3D-1

servicenow All Favorites History : DIAGNOSTIC TEST - Create Created Application scope: Medical Diagnosis Center Update set: Default [Medical Diagnosis Center]

DIAGNOSTIC TEST New record

Test Name Test Code

Category -- None -- Active ☐

Test Duration Price \$ 0.00

Description

Submit

5.4 Phase 4 — Appointment Table Development

Create the Appointment table to manage patient appointments.

Implementation

Create the Appointment table and configure the required fields. Establish the required relationship between the patient and diagnostic test so that an appointment can contain the selected patient and diagnostic test information.

Result

The Appointment table is created and appointment records can be maintained.

Figure 9: Appointment Table

servicenow All Favorites History : APPOINTMENT TABLE - Create Cre... Application scope: Medical Diagnosis Center Update set: Default [Medical Diagnosis Center]

APPOINTMENT TABLE New record

Appointment Number

Patient

Status -- None --

Approval Status -- None --

Email

Approval Date

Test

Scheduled Date

Slot Time Configurable time slots

Rejection Reason

Confirmation Sent ☐

Created By

Approved By

Notes

Payment Status -- None --

Total Amount \$ 0.00

Refund Amount \$ 0.00

Submit

5.5 Phase 5 — Report Table Development

Create the Report table to maintain diagnostic reports related to completed appointments.

Implementation

Create the Report table and configure the required fields for report management. The report record is associated with the relevant appointment so that the generated report can be maintained against the appointment.

Result

The Report table is created and report information can be managed.

Figure 10: Report Table

The screenshot displays the ServiceNow interface for creating a new record in the 'REPORT TABLE'. The form includes fields for Patient, Appointment, Report Date, Verified By, and Report File. A 'Findings' text area and a 'Submit' button are also present. The browser address bar shows the URL: dev425728.service-now.com/now/nav/ui/classic/params/target/x_2088302_medica_0_report_table.do%3Fsys_id%3D-1. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', and a 'REPORT TABLE - Create Created' button. The application scope is 'Medical Diagnosis Center' and the update set is 'Default [Medical Diagnosis Center]'.

5.6 Phase 6 — Business Rules and Automation

Configure the required business rules and automation for the Medical Diagnosis Center process.

Implementation

Configure the required automation for appointment management, appointment cancellation, refund calculation, and report generation after appointment completion. The configured rules perform the required actions based on the appointment process.

Result

The required business rules and automation are configured for the Medical Diagnosis Center application.

Figure 11: Business Rule and Automation Configuration

servicenow

AllFavoritesHistoryWorkspaces

Business Rule - Auto Populate Refund Amount Cancellation

Application scope: Medical Diagnosis Center

Update set Default (Medical Diagnosis Center)

UpdateDelete

Business Rule

Auto Populate Refund Amount Cancellation

A business rule is a server-side script that runs when a record is displayed, inserted, deleted, or when a table is queried. Use business rules to automatically change values in form fields when the specified conditions are met. [More Info](#)

Name

Auto Populate Refund Amount Cancellation

Table

APPOINTMENT TABLE [x_2088302_medic...

Application

Medical Diagnosis Center

Active

Advanced

When to run

Actions

Advanced

Condition

Script

Turn on ECMAScript 2021 (ES12) mode

```
1 (function executeRule(current, previous /*null when async*/) {
2   // Only process if status is 'cancelled' and total_amount exists
3   if (current.status == 'cancelled') {
4     var totalAmount = parseFloat(current.total_amount.toString()) || 0;
5     // Calculate refund amount with 10% deduction
6     // 10% deduction means customer gets 90% back
7     var deductionAmount = totalAmount * 0.10;
8     var refundAmount = totalAmount - deductionAmount;
9     // Set the refund amount
10    current.refund_amount = refundAmount.toFixed(2);
11  }
12 })(current, previous);
```

Update

Delete

Related Links

Run Point Scan

servicenow

AllFavoritesHistoryWorkspaces

Business Rule - Auto Create Report Appointment Complete

Application scope: Medical Diagnosis Center

Update set Default (Medical Diagnosis Center)

UpdateDelete

Business Rule

Auto Create Report Appointment Complete

A business rule is a server-side script that runs when a record is displayed, inserted, deleted, or when a table is queried. Use business rules to automatically change values in form fields when the specified conditions are met. [More Info](#)

Name

Auto Create Report Appointment Complete

Table

APPOINTMENT TABLE [x_2088302_medic...

Application

Medical Diagnosis Center

Active

Advanced

When to run

Actions

Advanced

Condition

Script

Turn on ECMAScript 2021 (ES12) mode

```
1 (function executeRule(current, previous /*null when async*/) {
2
3   // 1. Point to your verified Report Table
4   var grReport = new GlideRecord('x_2088302_medical_report_table');
5   grReport.initialize();
6
7   // 2. Map fields from Appointment to Report
8   grReport.setValue('appointment', current.sys_id);
9   grReport.setValue('patient', current.getValue('patient'));
10  grReport.setValue('verified_by', current.getValue('approved_by') || gs.getUserID());
11
12  // Set today's date
13  var gDate = new GlideDate();
14  grReport.setValue('report_date', gDate.getValue());
15
16  // Generate Report Number: RPT-(Appointment Number)
17  var rptNumber = current.getValue('appointment_number') || current.getValue('number') || '';
18  grReport.setValue('report_number', 'RPT-' + rptNumber);
19 }
```

Update

Delete

Related Links

Run Point Scan

servicenow

AllFavoritesHistoryWorkspacesAdmin

Business Rule - rejectionDetailsUpdate

Application scope: Medical Diagnosis Center

Update set Default (Medical Diagnosis Center)

UpdateDelete

Business Rule

rejectionDetailsUpdate

A business rule is a server-side script that runs when a record is displayed, inserted, deleted, or when a table is queried. Use business rules to automatically change values in form fields when the specified conditions are met. [More Info](#)

Name

rejectionDetailsUpdate

Table

APPOINTMENT TABLE [x_2088302_medic...

Application

Medical Diagnosis Center

Active

Advanced

When to run

Actions

Advanced

Condition

Script

Turn on ECMAScript 2021 (ES12) mode

```
1 (function executeRule(current, previous /*null when async*/) {
2   if (current.approval_status == "rejected") {
3     current.rejection_reason = "Cancelled/Rejected by the user";
4     current.notes = "";
5     current.setValue("approved_by", null);
6   }
7   if (current.approval_status == "approved") {
8     current.notes = "Approved by the System Admin";
9     current.approved_by = gs.getUserID();
10    current.rejection_reason = "";
11  }
12 })(current, previous);
```

Update

Delete

Related Links

Run Point Scan

5.7 Phase 7 — Service Portal Development

Develop the Service Portal to provide users with access to the Medical Diagnosis Center services.

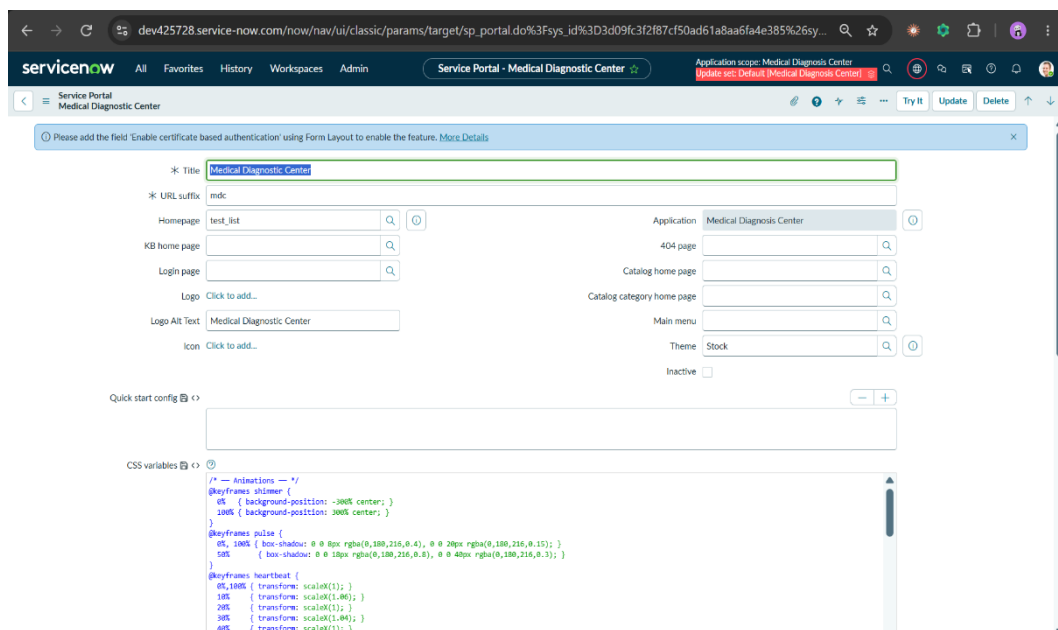
Implementation

Create and configure the Service Portal and provide the required portal components for accessing the Medical Diagnosis Center functionality. The portal provides a simple interface for users to interact with the application.

Result

The Medical Diagnosis Center Service Portal is created and accessible to users.

Figure 12: Medical Diagnosis Center Service Portal



5.8 Phase 8 — Portal Pages and Widgets

Configure the required pages and widgets for the Medical Diagnosis Center Service Portal.

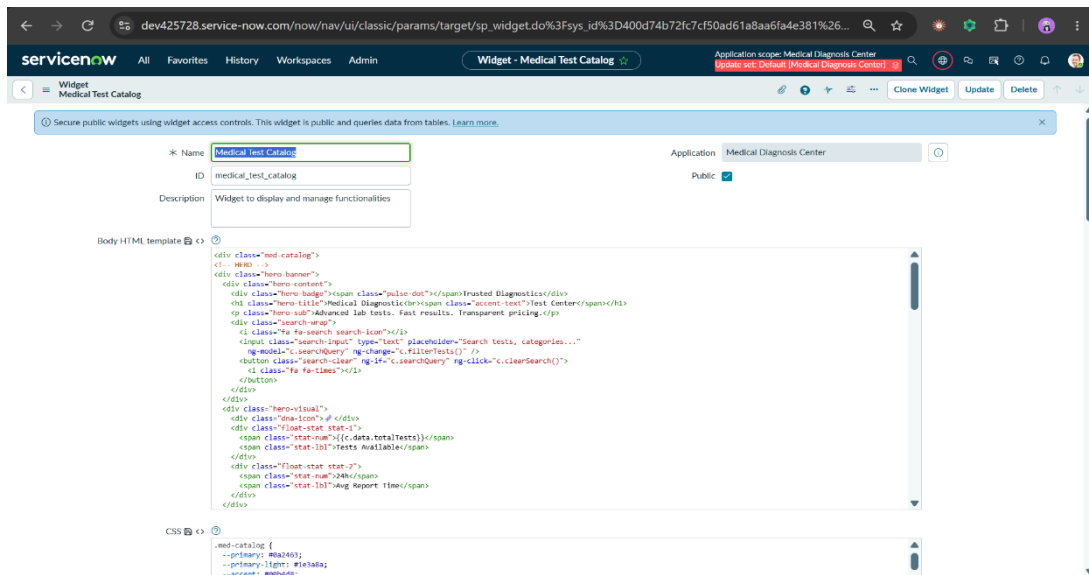
Implementation

Create or configure the required portal pages and widgets for the application. The widgets provide the user interface required for accessing and performing the available Medical Diagnosis Center functions.

Result

The required Service Portal pages and widgets are configured and available to users.

Figure 13: Service Portal Pages and Widgets



5.9 Phase 9 — Appointment Booking

Implement the appointment booking process through the Medical Diagnosis Center application.

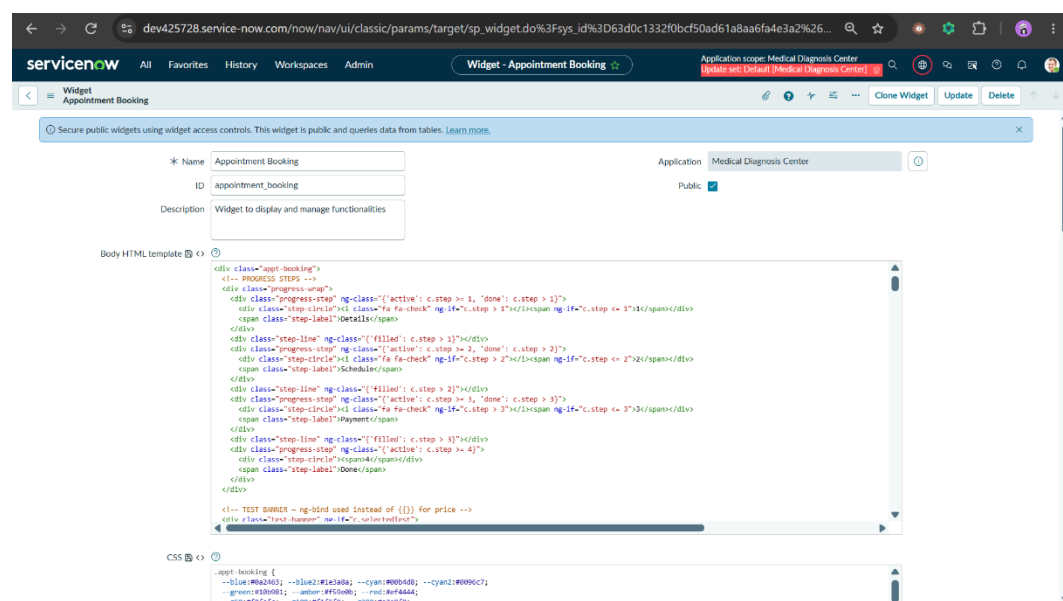
Implementation

Select the required patient and diagnostic test and enter the required appointment information. Submit the appointment details to create the appointment record in ServiceNow.

Result

The appointment is created successfully and associated with the required patient and diagnostic test.

Figure 14: Appointment Booking



5.10 Phase 10 — Appointment Cancellation and Refund

Implement appointment cancellation and refund calculation for cancelled appointments.

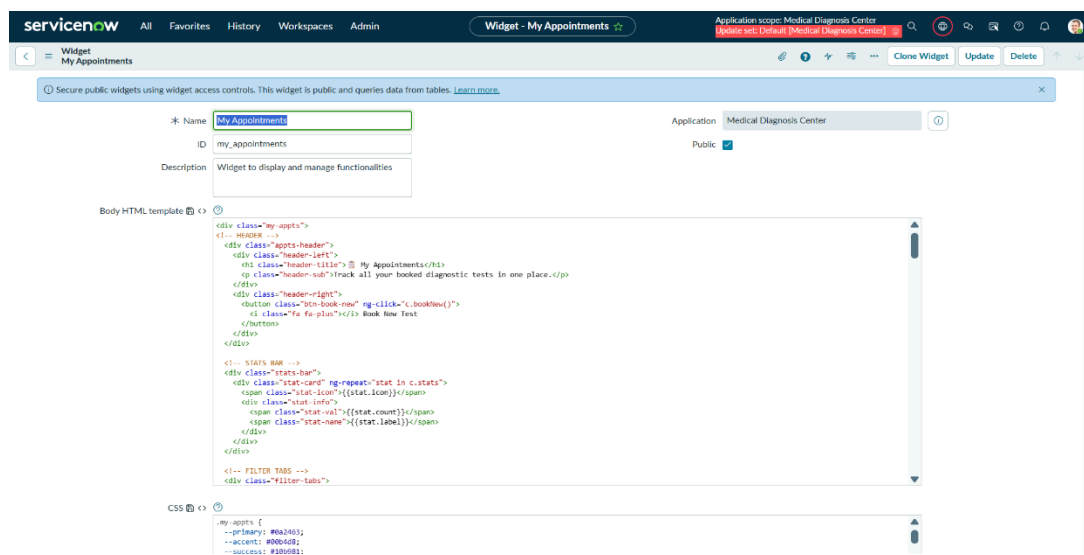
Implementation

Open the required appointment and perform the cancellation process. The configured system logic calculates the applicable refund for the cancelled appointment according to the project requirements.

Result

The appointment is cancelled and the required refund amount is calculated.

Figure 15: Appointment Cancellation and Refund Calculation



5.11 Phase 11 — Appointment Completion and Report Generation

Implement automatic report generation after completion of an appointment.

Implementation

Update the appointment to the completed state. The configured automation identifies the completed appointment and generates the corresponding report record.

Result

The diagnostic report is automatically generated for the completed appointment.

Figure 16: Automatic Report Generation

[Insert screenshot showing the generated report here]

5.12 Phase 12 — Report Management

Manage the reports generated for completed appointments.

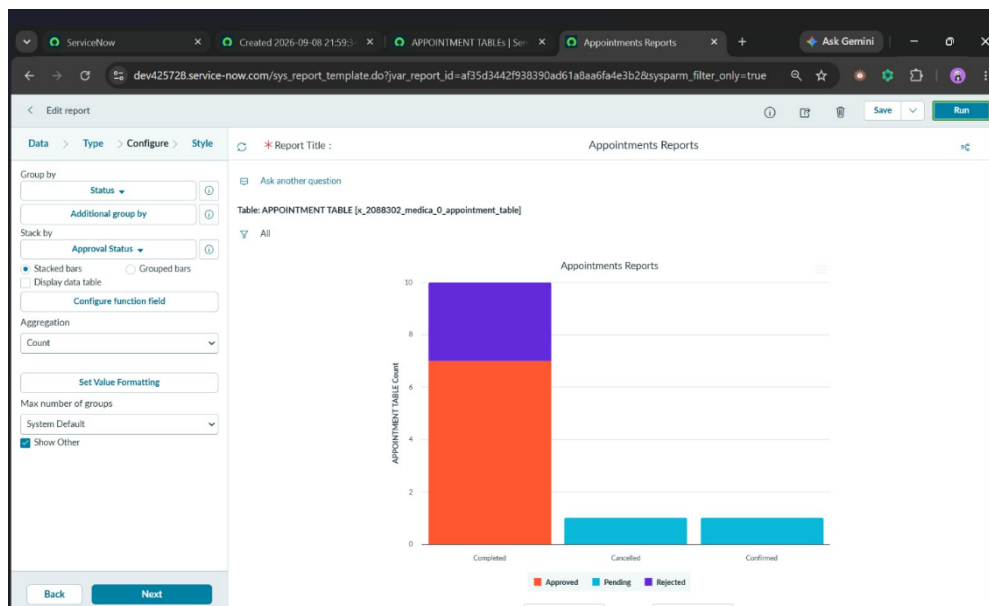
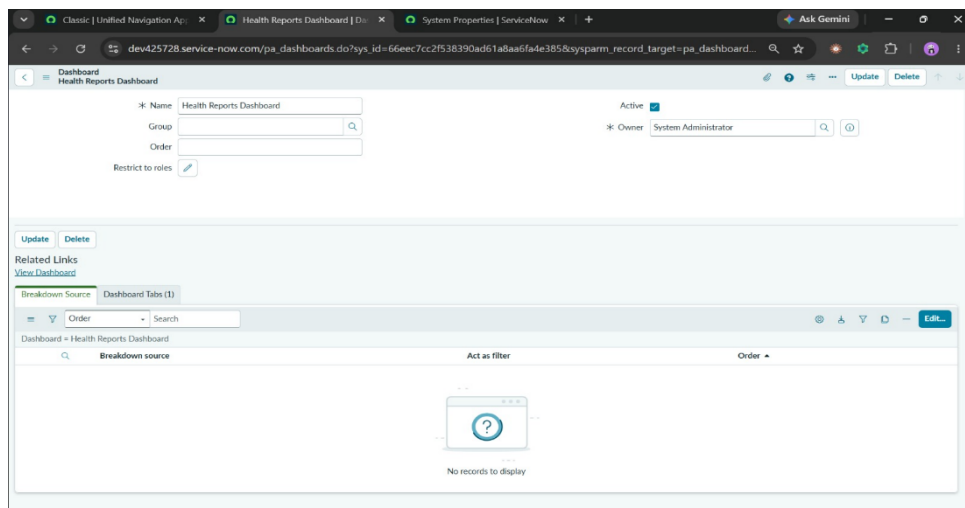
Implementation

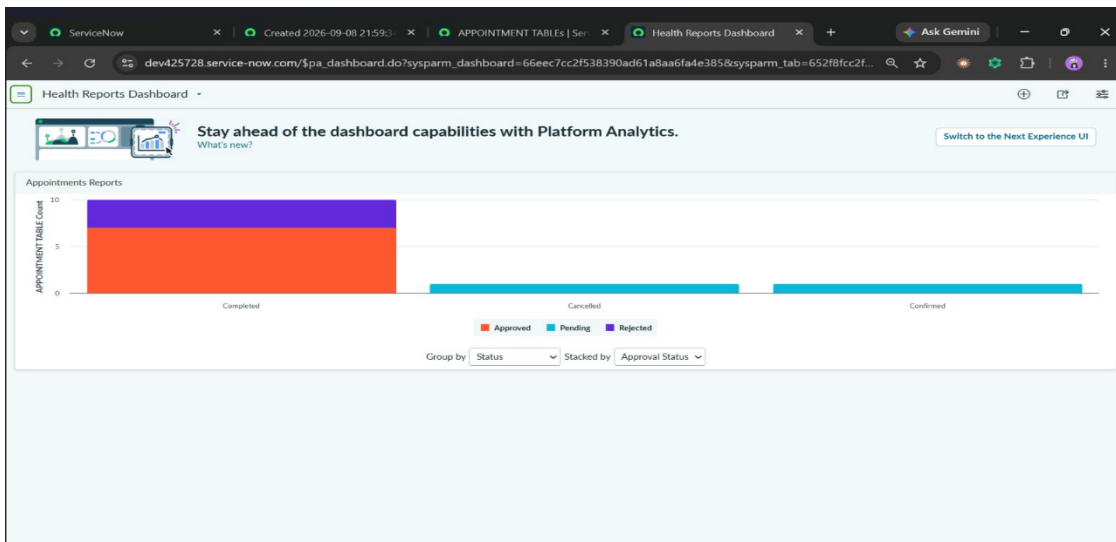
Open the generated report and verify the associated appointment and required report information. The report record can be maintained through the ServiceNow application.

Result

The generated report is available and associated with the relevant appointment.

Figure 17: Report Management





5.13 Phase 13 — Final Validation

The completed solution is checked to confirm that the main records and processes are working correctly.

- Patient registration is available.
- Diagnostic tests can be maintained and selected.
- Appointments can be created and managed.
- Appointment cancellation is supported.
- Refund calculation is performed for cancelled appointments.
- Completed appointments generate the required report.
- Reports can be managed.
- Service Portal access is available.

5.14 Phase 14 — Testing and Deployment Validation

The Medical Diagnosis Center application is tested end-to-end to confirm that the implemented activities work together as planned.

Implementation

- Create and manage patient records.
- Verify diagnostic test records and selection.
- Verify appointment booking and management.
- Verify appointment cancellation and refund calculation.

- Verify appointment completion and report generation.
- Verify report management.
- Verify Service Portal functionality.

5.15 Conclusion

The project demonstrates a Medical Diagnosis Center application developed using ServiceNow Service Portal. The implemented solution supports patient registration, diagnostic test management, appointment booking and management, appointment cancellation, refund calculation, automatic report generation, report management, and Service Portal access.